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CareerPilot

Live Application URL: [CareerPilot](#)

Tech Stack: [CareerPilot Tech Stack](#)

System Architecture Document: [CareerPilot System Design](#)

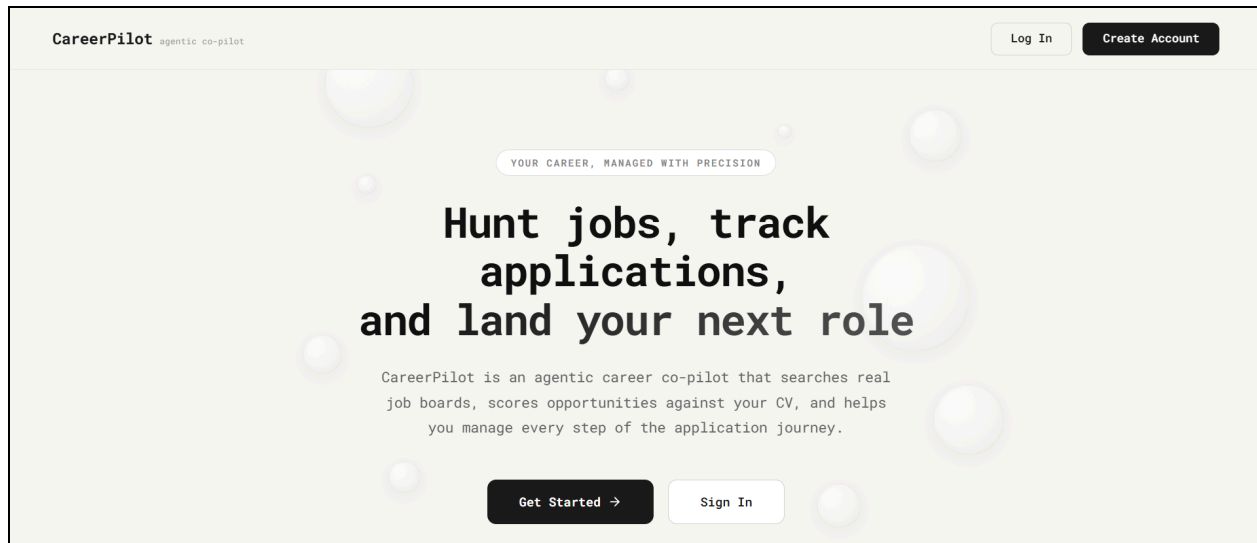


Figure : Landing Page of CareerPilot

CareerPilot is an AI-powered career assistance platform designed to help job seekers manage their career journey through intelligent automation, personalized recommendations, and productivity tracking.

The platform leverages **Retrieval-Augmented Generation (RAG)**, Artificial Intelligence, and modern web technologies to provide users with personalized career guidance based on their resumes, skills, experiences, and career goals.

Unlike traditional job portals, CareerPilot acts as a personal career assistant that understands the user's background, recommends suitable jobs, identifies skill gaps, generates personalized cover letters, and tracks career progress in a unified platform.

Modern job seekers face several challenges:

- Job opportunities are scattered across multiple platforms.
- Generic AI tools lack personalization.
- Users struggle to identify skill gaps.
- Application tracking is often manual.
- Career planning and learning roadmaps are difficult to manage.

CareerPilot addresses these issues by creating a centralized AI-powered ecosystem that understands the user and actively assists throughout the job search process.

The primary objectives of CareerPilot are:

- Build a personalized AI career assistant.
- Match users with relevant job opportunities.
- Analyze resumes and identify strengths and weaknesses.
- Generate AI-assisted cover letters.
- Provide skill gap analysis and learning roadmaps.
- Track applications, goals, and career progress.
- Maintain conversational memory through RAG.

Core features of this system includes:

Authentication System

Users can:

- Register accounts
- Login securely
- Manage profile information
- Access personalized dashboards

Resume Intelligence (RAG Core)

Workflow	Benefits
● User uploads resume in (PDF or DOCX or DOC format). ● Resume text is extracted. ● Text is divided into meaningful chunks (e.g. Skills, Education, Experience and Projects). ● Chunks are converted into vector embeddings. ● Embeddings are stored in ChromaDB. ● AI retrieves relevant chunks when answering queries.	● Personalized responses ● Reduced hallucination ● Accurate job matching ● Context-aware recommendations

Job Hunter Agent

Features	Output
• Job searching • Personalized filtering • Career-based recommendations • Resume-aware matching	• Job Title • Company Name • Location • Required Skills • Match Percentage • Recommendation Explanation

Fit Score Engine

CareerPilot computes a compatibility score between a user's resume and a job description.

Evaluation parameters	Output
• Technical Skills Match • Project Experience • Educational Background • Relevant Technologies • Experience Level	• Match Percentage • Missing Skills • Strength Areas • Improvement Suggestions

AI Career Assistant

A conversational assistant that understands the user's career profile. Supported actions and queries -

- **Resume Analysis** - Review my CV, What are my strengths?
- **Skill Gap Analysis** - What skills am I missing for this role? How can I improve my profile?
- **Career Guidance** - Which jobs suit me? What should I learn next?
- **Cover Letter Generation** - Generate personalized cover letters, Tailor applications for specific positions.
- **Learning Roadmaps** - Create weekly learning plans, Recommend resources and certifications according to Resume.

Productivity & Goal Tracking

Goal management	To-do management	Deadline tracking
• Apply 1 to 10 jobs this week • Complete React course • Update resume	• Create tasks • Update status • Track completion	• Application deadlines • Learning milestones • Weekly goals

Application Tracker

A Kanban-style application management system.

Columns	Features
• Applied • Interviewing • Offer • Rejected	• Drag-and-drop status updates • Application history • Progress monitoring

Dashboard & Analytics

The dashboard provides real-time insights & AI nudge.

Statistics	Benefits
• Total Applications • Interviews Scheduled • Offers Received • Goals Completed • Learning Progress	• Progress visibility • Career performance tracking • Motivation through measurable growth

Run This Project

View Readme for detailed explanation : [CareerPilot Readme](#)

Test Cases : [CareerPilot Test Cases](#)

To install django modules dependencies run the **requirements.txt** file in the server folder. Then run this in terminal :

```
pip install -r requirements.txt
```

Security Considerations

- Secure authentication
- Protected API endpoints
- Resume data isolation
- Server-side validation
- Controlled AI access

Future Enhancements

- LinkedIn integration
- Email notifications
- Interview preparation assistant
- Multi-language support
- Real-time recruiter matching
- Advanced analytics dashboard

Challenges Faced

- Resume parsing inconsistencies
- Maintaining RAG accuracy
- Job matching optimization
- Ensuring personalized AI responses
- Managing vector storage efficiently

CareerPilot transforms traditional job searching into an intelligent personalized experience by combining RAG-based AI technologies, the platform provides users with career guidance, job recommendations, application tracking, and productivity management in one centralized system. The project demonstrates how modern AI systems can be grounded in user-specific data to provide accurate, context-aware, and actionable career assistance.